

AWS Container & Serverless Services

EKS vs ECS vs Lambda — Feature & Cost Comparison

SERVICE OVERVIEW

Kubernetes-native

Amazon EKS

Managed Kubernetes — run pods on EC2 or Fargate

- ✓ Full Kubernetes API — pods, deployments, namespaces
- ✓ Helm Charts & ArgoCD / GitOps workflows
- ✓ EC2 node groups or serverless Fargate nodes
- ✓ Advanced Ingress, SSL/TLS, DNS, VPC networking
- ✗ \$0.10/hr control plane fee — always charged
- ✗ Steeper learning curve; more operational overhead

\$0.10 / hr

+ EC2 or Fargate node costs on top

AWS-native

Amazon ECS

AWS-native container orchestration — Tasks & Services

- ✓ Free control plane — pay compute only
- ✓ EC2 or Fargate launch types
- ✓ Deep AWS service integration (ALB, IAM, CloudWatch)
- ✓ Simple to set up and operate
- ✗ No Kubernetes, Helm, or ArgoCD support
- ✗ AWS vendor lock-in — not portable

Free

Pay EC2 or Fargate compute only

Serverless functions

AWS Lambda

Serverless functions — no servers or containers to manage

- ✓ 1M requests/month included in free tier
- ✓ Pay per millisecond of execution only
- ✓ Auto-scales to zero — no idle cost
- ✓ Zero infrastructure management
- ✗ 15-minute maximum execution time
- ✗ Cold starts can add latency

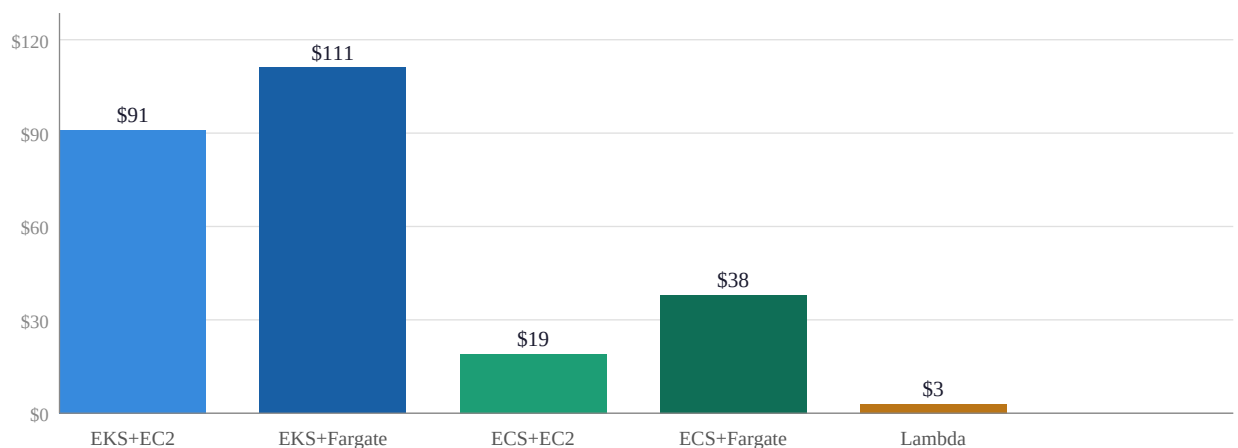
\$0.20 / 1M req

+ \$0.0000167 per GB-second compute

MONTHLY COST ESTIMATE

Based on 1 vCPU · 2 GB RAM · 730 hours/month (us-east-1 pricing, approximate)

■ EKS + EC2 ■ EKS + Fargate ■ ECS + EC2 ■ ECS + Fargate ■ Lambda



FEATURE COMPARISON

Feature	EKS	ECS	Lambda
Kubernetes API	✓ Full	✗ None	✗ None
Helm Charts	✓ Native	✗ No	✗ No
ArgoCD / GitOps	✓ Yes	✗ No	✗ No
Fargate (serverless)	✓ Yes	✓ Yes	✓ Always
Control plane cost	\$0.10/hr	Free	Free
Auto-scale to zero	~ With work	~ With work	✓ Native
Max execution time	Unlimited	Unlimited	15 min
Cold starts	✗ No	✗ No	~ Sometimes
Multi-cloud portable	✓ Yes	✗ AWS only	✗ AWS only
Best for	Microservices GitOps teams	Simple apps AWS-only	Event triggers Short tasks

WHEN TO USE EACH SERVICE

Use EKS when...	Use ECS when...	Use Lambda when...
You need Helm, ArgoCD, or GitOps workflows. Your team knows Kubernetes. You want multi-cloud portability.	You want simplicity and deep AWS integration without Kubernetes complexity. Cost efficiency matters. You are AWS-only.	You have event-driven, short-lived workloads: API triggers, S3 events, scheduled jobs. Traffic is unpredictable or near zero most of the time.

PRICING DETAIL (us-east-1)

Setup	Control Plane	Compute	Est. Monthly
EKS + EC2 (t3.small)	\$0.10/hr (\$72/mo)	~\$17–19/mo (t3.small)	~\$89–91
EKS + Fargate	\$0.10/hr (\$72/mo)	\$0.04048/vCPU/hr + \$0.004445/GB/hr	~\$109–111
ECS + EC2 (t3.small)	Free	~\$17–19/mo (t3.small)	~\$17–19
ECS + Fargate	Free	\$0.04048/vCPU/hr + \$0.004445/GB/hr	~\$36–38
AWS Lambda	Free	\$0.20/1M req + \$0.0000167/GB-sec	~\$0–5 (low scale)

* Prices are approximate and based on AWS us-east-1 region as of 2025. Actual costs vary by region, reserved instance discounts, data transfer, and usage patterns. Always check the AWS Pricing Calculator for accurate estimates.